GlucoTwin - Digital Twin architecture (Type 2 Diabetes, 2-hour glucose-excursion forecast)

1 DATA STREAMS (synthetic)

Static / historical stream

Synthetic EHR (Synthea-style)
Demographics, BMI, HbA1c, fasting glucose
BP, LDL, TG, eGFR, ICD-10 diagnoses
TCF7L2 genetic risk alleles, family history
Medication: metformin, sulfonylurea

Dynamic / real-time stream

Simulated CGM glucose (5-min)
Heart rate and HRV (RMSSD)
Steps / activity
Sleep stages (awake, light, deep, REM)
Mimics Apple Health / Google Fit / CGM feeds

Patient behaviour (hidden)

Indian meal patterns, stress events,
sleep debt, walking habits drive physiology

2 FUSION

Feature fusion

Leak-free rolling windows:
glucose level, slope, variability
24 h time-in-range
same clock-time, prior days
steps, HR excess, HRV drop
24 h sleep quality
EHR encoder (16 fields)

= 44-feature fused vector
sampled every 15 min

3 TWIN CORE

Forecast model (ML)

Gradient-boosted trees
P(glucose > 180 mg/dL in next 2 h)
Isotonic calibration + alert threshold
Expected 2 h peak (regressor)
Group-level explanations

Mechanistic simulator

Meal response, insulin resistance,
sleep debt, stress, exercise
Counter-factual 'what-if' runs
with paired meals and noise

Twin state (VirtualPatient)

EHR + rolling wearable history
forecast(t) · what_if(change)

4 CLINICIAN VIEW

Doctor dashboard

Streamlit app

Risk level + probability
Live CGM + 2 h window
Why? (driver bars)
HR / HRV / steps panel
What-if simulator
Patient EHR card

Alert

Early warning up to
2 h ahead of the
excursion, so the
patient / care team
can act

Synthetic data only · patient-level train/val/test split · DPDP-safe · MIT licence